Matheus Hoffmann

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Profile

Engineer who has built forecasting systems that run in production since 2017, and the data and MLOps platforms that keep them running: loss forecasting for an e-commerce fraud platform, churn forecasting for a payment company, audience forecasting for a broadcaster. More recently that practice turned to the sea: a bit-for-bit GPU port of the WAVEWATCH III wave model, an open platform that turns public sea, weather and bathing-water data into a daily signal for three Brazilian coastal cities, and a proposal for a national benchmark that scores wave forecasts against instruments. Before that, two years in an ocean instrumentation laboratory at UFRJ. What I offer an ocean-research group: numerical wave modelling on GPUs, operational forecasting pipelines with verification built in, data infrastructure that gathers and standardises observations from many sources, and an open-science practice: each work below has a DOI, a public repository and a pinned, reproducible toolchain.

Research

marola · Hoffmann, M., Valério, B., Soares da Silva Junior, R. K., Oliveira, E., Almeida, L. R. and Ribeiro, P. (2026). *marola: an open, non-profit platform for sea conditions and bathing-water quality at Brazilian beaches, built on public data.* Software, v0.2.2. DOI 10.5281/zenodo.23224155 GitHub marola-dev/marola
Scores the beaches of Florianópolis, Rio de Janeiro and Salvador for today and tomorrow from sea, wind, wave and tide forecasts and the official bathing-water bulletins of three state agencies; plain rules, not a model, decide anything that concerns a swimmer's safety. Six authors, in three countries.

ww3-gpu · Hoffmann, M. (2026). *WW3 GPU Lab: agentic coding and agentic research on a bit-for-bit GPU port of WAVEWATCH III.* Software, v0.1.3. DOI 10.5281/zenodo.23221351 GitHub h0ffmann/ww3-gpu
Moves the expensive kernels of the WAVEWATCH III third-generation spectral wave model to GPUs without changing the answer: a C++/Kokkos port of the nonlinear four-wave interaction term (W3SNL1, DIA) shown bit-identical to the Fortran on the Serial, OpenMP and CUDA backends, with CPU (OpenMP, MPI) and GPU (OpenACC, CUDA Fortran) benchmarks that each carry the command that reproduces them.

marola-bench · Hoffmann, M. (2026). *marola-bench: a proposal for a WeatherBench for Brazil.* Proposal, v0.1.0. DOI 10.5281/zenodo.23352633 GitHub marola-dev/marola-research
Proposes a public benchmark for forecasts on the Brazilian coast in the WeatherBench tradition: a fixed region, scores for ocean waves, and in-situ instruments as the ground truth, so that any forecaster can be scored at the same observation points over time.

Also research on open-source software: *Closed Source: The God that Failed* (DOI 10.5281/zenodo.23333656) | personal website: h0ffmann.github.io/h0ffmann
Other works with a DOI: *picasso*, the Nix flakes that pin the toolchains the works above build with (10.5281/zenodo.23311208); *Nashkell: Rhizomatic experiments*, an open book (10.5281/zenodo.23356739).

Experience

Senior Machine Learning Operations (MLOps) Engineer, TXP / ITV

July 2025 – August 2026 · London, UK (remote)

- Re-architected a single-channel audience-forecasting pipeline into an extensible multi-channel ML platform, extending the forecasts to ITV2, ITV3 and ITV Quiz
- Owned the pipeline end to end, from data ingestion to the published forecasts: AWS Glue ETL, Amazon SageMaker training and MLflow tracking, orchestrated on Apache Airflow (MWAA), served from Amazon DynamoDB

Senior ML Engineer II (L5), Signifyd (via the Legiti acquisition, March 2023)

June 2022 – November 2024 · San Jose, CA, USA (remote)

- Loss forecasting: cut model experimentation time by 93% (16 hours to 1 hour) by refactoring the simulation code, adding multithreading and moving the pandas logic to PySpark
- Cut batch feature computation time and cost by 77% by moving it from PostgreSQL to PySpark on Databricks
- Built anomaly detection for fraud risk analysis on Databricks (Unity Catalog, Delta Lake, MLflow, PySpark MLlib)

Lead MLOps Engineer, Elemeno AI

June 2021 – July 2022 · San Francisco, CA, USA (remote)

- Designed and built a full open-source MLOps stack on Kubeflow (KServe, Katib), MinIO, ArgoCD and GitLab CI on Google Kubernetes Engine, provisioned with Terraform and Helm, for Americanas (BVMF: AMER3): in effect an open-source distribution of what managed platforms such as the early Vertex AI offered
- Gave data scientists a self-service experimentation platform (JupyterHub, Elyra, MLflow); three open-source repositories published at github.com/elemeno-ai

Senior Software Engineer, Broad (Y Combinator, W21)

November 2020 – June 2021 · London, UK (remote)

- Built the single source of truth for banking events in Scala, ZIO, Event Sourcing and Kafka Streams, integrating BIN-sponsorship and foreign-exchange providers

Software Engineer, iU Pay / Guiabolso

May 2019 – November 2020 · São Paulo, Brazil (remote)

- Core contributor to Brazil's first instant cross-bank transaction engine, delivered eight months before the public launch of Pix: the transaction engine, the clearing process and the Apache Cassandra data model
- Processed transactions from Apache Kafka with Event Sourcing and RPA; stayed on through the Guiabolso acquisition

Software Developer, Stone (Nasdaq: STNE)

October 2017 – April 2019 · Rio de Janeiro, Brazil (remote)

- Churn forecasting: improved the F1-score and the training speed of the churn-prediction models by migrating legacy Bayesian networks to XGBoost/LightGBM on Apache Spark
- Cut the logistics cost of switching mobile carriers on point-of-sale terminals by 35% with a recommendation system based on signal quality and geographic proximity

Student Researcher, LIOC – Ocean Instrumentation Laboratory, COPPE/UFRJ

September 2015 – October 2017 · Rio de Janeiro, Brazil

- FPGA signal processing in LabVIEW (NI cRIO-9068) for a Brazilian Navy underwater-communication programme, and real-time GPS tracking of boats in competitions on the TI MSP430 with Python, PHP and MySQL

Skills

- **Machine Learning Operations (MLOps)**: end-to-end ML pipelines, model training and deployment, experiment tracking (MLflow), Kubeflow, Amazon SageMaker, feature engineering
- **Data Engineering**: Apache Spark, PySpark, Databricks, Delta Lake, Apache Airflow, dbt, ETL/ELT, SQL
- **Cloud & Infrastructure**: Amazon Web Services (AWS: S3, EC2, EKS, EMR, Kinesis, Glue, DynamoDB, Redshift), Google Cloud (GKE), Kubernetes, Docker, Terraform, Helm, ArgoCD, CI/CD (GitHub Actions, GitLab CI, Jenkins)
- **Backend & Distributed Systems**: Scala, Python, TypeScript, FastAPI, REST APIs, Apache Kafka, Event Sourcing, Akka, ZIO, PostgreSQL, Apache Cassandra
- **LLM & AI Engineering**: RAG, Model Context Protocol (MCP), DSPy, fine-tuning (QLoRA, DPO), Claude Code

Education

Bachelor's in Electronics and Computer Engineering, Federal University of Rio de Janeiro (UFRJ)

2013 – present · Rio de Janeiro, Brazil

Final project in progress on the GPU optimisation of the WAVEWATCH III ocean wave model with LabECO/UFSC (github.com/h0ffmann/ww3-gpu). Key topics: software engineering, big data and HPC.

Technician in Electronics, Federal Center of Technological Education CEFET/RJ

2010 – 2012 · Rio de Janeiro, Brazil

Industrial electronics and instrumentation.

Languages, certifications and honours

English (C1, fluent), Portuguese (native), Spanish (reading).

AWS Cloud Practitioner · AWS ML Specialty · AWS ML Engineer Associate · AWS Data Engineer Associate · PADI Rescue Diver

Oceanman Rio de Janeiro, 3rd place in the 30–39 age group, 10 km open-water swim (September 2025). Signifyd "Values VIP: Tenacious" award for technical excellence, voted by employees (February 2024).

References available on request.